

SE – LAB 1

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PROBLEM STATEMENT : 13- Patient Health Record Consent Management System

Use-Case Specification: Grant Granular Consent

Document Version: 1.0

Use Case ID: UC-01

Use Case Name: Grant Granular Consent

Primary Actor: Patient

Secondary Actor / Service: Consulting Doctor / Diagnostic Clinic, Audit Logger

1. Brief Description

Enables an authenticated patient to select a verified healthcare practitioner or clinic, specify granular access permissions for specific diagnostic records, set a time-bound validity window, and issue a cryptographically verifiable access token.

2. Preconditions

1. The Patient is successfully authenticated and logged into the Patient Health Record portal.
2. The target Consulting Doctor or Clinic has been verified and registered by the Clinic Administrator.
3. The Patient has valid medical records/diagnostic reports stored in the system gateway.

3. Postconditions

- **Success (Guaranteed):**
 - A time-bounded consent token with granular record scopes is generated and active.
 - An immutable audit record detailing the grant event (Patient ID, Doctor ID, scope, timestamps) is permanently committed to the audit trail.
 - The patient's consent management dashboard is updated in real time.

- **Failure:**
 - No access token is issued, no data access permissions are altered, and the error state is recorded in system logs.

4. Main Success Scenario (MSS)

1. Patient navigates to the **Consent Management** module and selects **"Grant New Consent"**.
2. System fetches and displays a searchable directory of verified clinics and consulting doctors.
3. Patient searches for and selects the intended consulting doctor.
4. System displays the patient's categorized medical record types (e.g., Blood Panel, Radiology/MRI, Prescription History).
5. Patient selects the specific record categories to share (excluding all unselected records).
6. Patient defines the access duration window (e.g., standard default 24 hours or custom duration).
7. Patient reviews the consent summary and clicks **"Authorize Access"**.
8. System executes **UC-06: Log Audit Trail** to write the complete transaction to the append-only log.
9. System generates an authorized, time-bounded JWT/consent token mapped to the doctor's credentials.
10. System displays a confirmation receipt: *"Consent successfully granted until [Expiry Timestamp]"* and redirects the patient to the active dashboard.
11. Use case ends successfully.

5. Alternate Flows

- **3a. Selected Doctor/Clinic Is Suspended or Unverified:**
 - **3a1.** System checks provider verification status and finds it inactive or revoked by the Clinic Admin.
 - **3a2.** System displays an error alert: *"The selected provider's credentials are not currently verified. Access cannot be granted."*
 - **3a3.** Patient selects an alternative verified doctor or cancels the operation.
 - **3a4.** Flow resumes at Step 3 or terminates.
- **6a. Invalid Custom Duration Window:**
 - **6a1.** Patient inputs an expiration timestamp in the past or exceeds the maximum allowable duration (e.g., > 30 days).
 - **6a2.** System triggers an inline validation alert: *"Invalid duration. Consent window must be between 1 hour and 30 days."*
 - **6a3.** Patient corrects the duration.
 - **6a4.** Flow resumes at Step 7.
- **8a. Audit Log Service Failure:**

- **8a1.** Append-only audit logger encounters a connection timeout or write failure.
- **8a2.** System halts token generation immediately to prevent unlogged access grants (Fail-Secure policy).
- **8a3.** System rolls back the transaction and displays: *"Authorization failed due to system security check. Please try again."*
- **8a4.** Use case terminates in a failure state.